

Cheng Na

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EDUCATION

Huazhong University of Science and Technology (985 & double first-class) Sept. 2024 – Present
Master of Engineering | Software Engineering | GPA: 3.78/4.00

- **Core modules:** Industrial Software Development (93/100), Software Process and Project Management (92/100)
- **Honors:** Postgraduate Scholarship (twice); Research and Innovation Scholarship

Shandong University (985 & double first-class) Sept. 2020 – Jun. 2024
Bachelor of Engineering | Software Engineering | GPA: 3.29/4.00

- **Core modules:** Operating System (95/100), Software Quality Assurance and Testing Technology (95/100)
- **Honors:** Outstanding Individual Award

RESEARCH OUTPUTS

- **A CVAE-enhanced generative adversarial network for end-to-end designs of auxetic metamaterials using NURBS control-point density representations.** (*Composite Structures, IF=7.8, CAS Q1 Top*)
Na Cheng, Hua Cao, Wei Luo, Xiongbing Fang, Liang Gao, Jie Gao.
- **A generative design method for auxetic metamaterials based on NURBS control-point density representations.** Inventors: Jie Gao, Na Cheng, Hua Cao et al. (*patent pending*)

RESEARCH & PROJECT EXPERIENCE

NURBS-Based Generative Inverse Design of Auxetic Metamaterials Jun. 2025 – Feb. 2026

- Built a dataset of 2d auxetic metamaterial unit cells from scratch using NURBS control-point representations.
- Trained a CVAE on the limited-sample dataset and performed latent-space interpolation to generate smooth structural transitions for data augmentation.
- Trained a ResNet-based property predictor and a CWGAN to generate structures conditioned on target properties.

Generative Inverse Design with Neural Operators and Diffusion Models May 2026 – Present

- Developed a property-conditioned diffusion model incorporating elastic-tensor embeddings, FiLM-conditioned residual blocks, self-conditioning, and multi-scale self-attention.
- Explored neural operators for forward prediction of structural mechanical properties, balancing inference efficiency with physical fidelity.
- Applied classifier-free guidance (CFG) for controllable property-conditioned generation and explored Universal Guidance for preference-guided generation.

Generative Design for Editable CAD Models Oct. 2024 – Present

- Participated in the KGJ Key Project (2025–2027), targeting the on-demand generation of editable CAD models and investigating CAD sequences and CadQuery programs as data representations.
- Used CATIA to model the target structures and expand the CAD model dataset required for the project.
- Developed a Transformer-based sequence generation model that semantically encodes design descriptions and autoregressively generates CAD modeling sequences.

Multi-Tool Agent for Medical Image Analysis Mar. 2026 – Jul. 2026

- Reimplemented and extended the MedRAX agent using LangGraph, enabling query interpretation, medical-imaging tool invocation, multi-tool result integration, and response generation.
- Added on-demand loading of tool models, medical RAG, long-term memory, and context management to support multi-turn medical-image analysis.

COMPETITION AWARDS

- **National Third Prize**, the 22nd China Postgraduate Mathematical Contest in Modeling, 2025
- **National Third Prize**, the 21st China Postgraduate Mathematical Contest in Modeling, 2024
- **Provincial Third Prize**, C/C++ Track, the 16th Lanqiao Cup, 2025

SKILLS

- **Languages:** Mandarin (Native), English (CET-6: 553)
- **Programming Languages:** Python (Advanced), Java (Intermediate), C/C++ (Familiar)
- **Research and Proposal Writing:** Experienced in drafting research proposals, project applications, technical reports, and academic manuscripts. Contributed to writing multiple project applications.